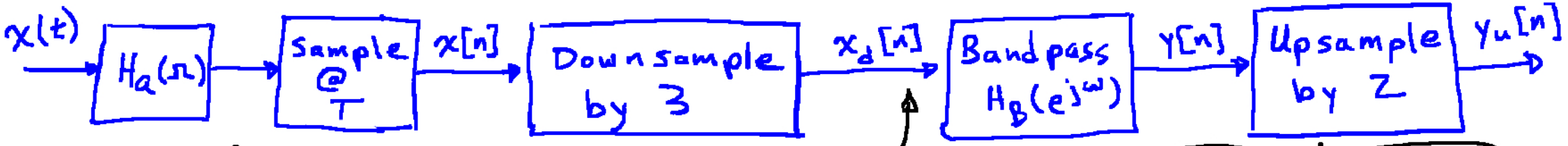
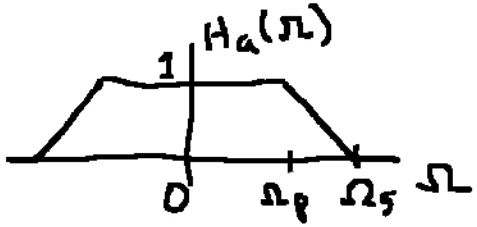
Upsampling and Downsampling Example

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Video Screenshots

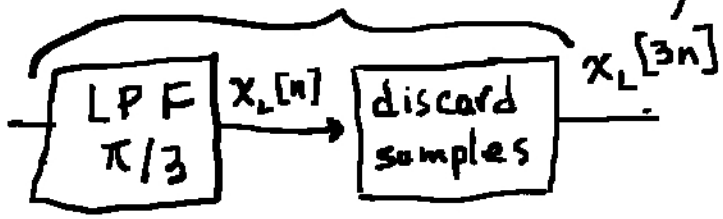


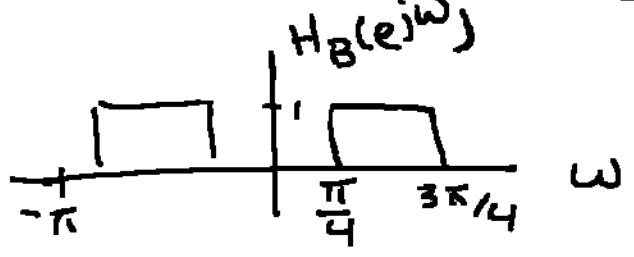


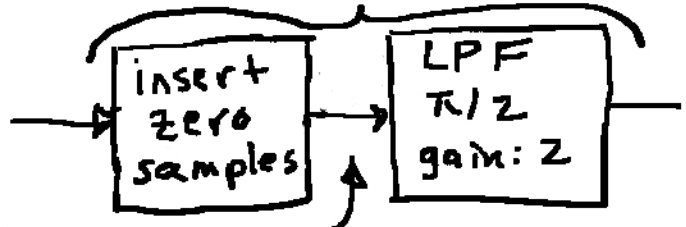
$\Omega_p = 1500\pi$ rads/sec

$\Omega_s = 3000\pi$ rads/sec

$T = \frac{1}{3000}$ sec

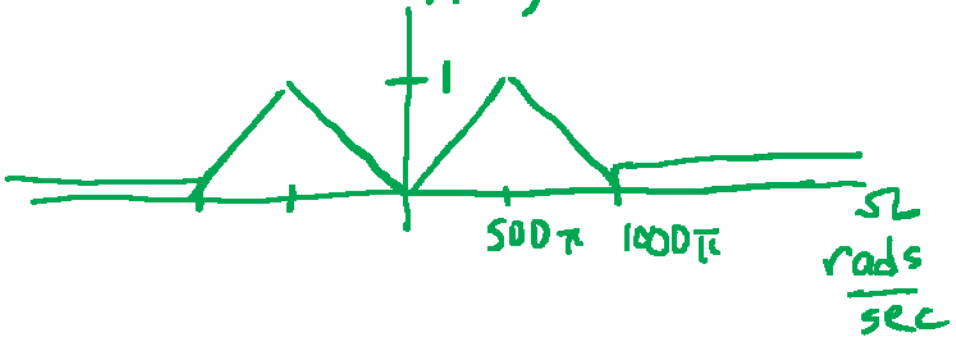




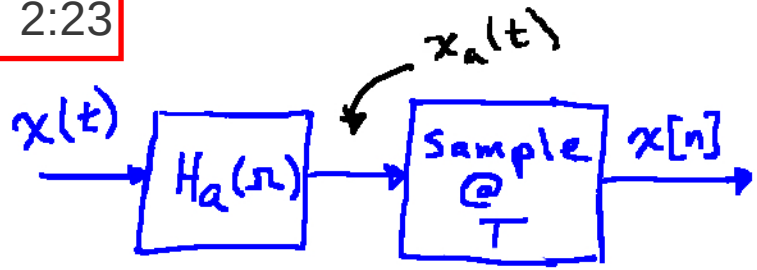


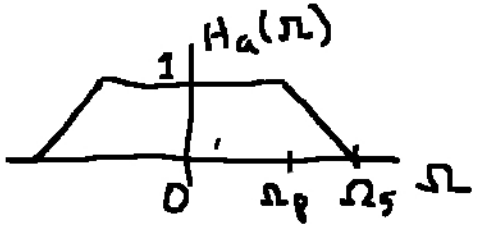
$$y_z[n] = \begin{cases} y[n/2] & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

Suppose



Find DTFT and FT representations for $x[n], x_d[n], y[n], y_u[n]$



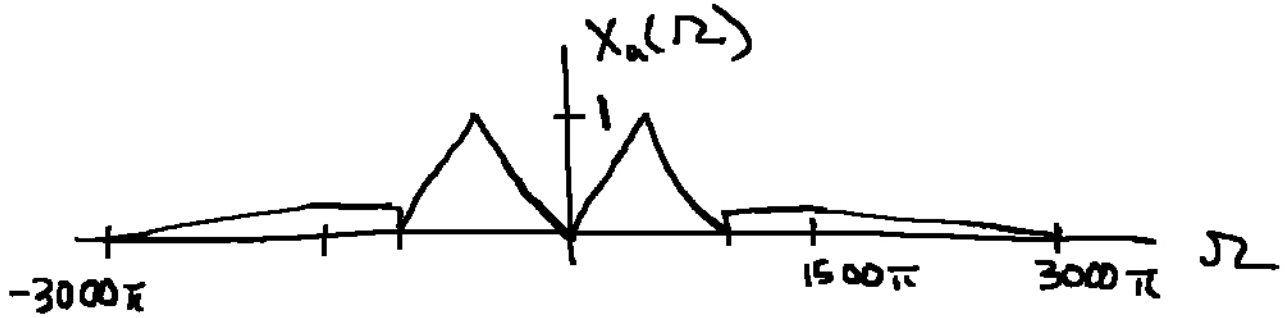


$\Omega_p = 1500\pi$ rads/sec

$\Omega_s = 3000\pi$ rads/sec

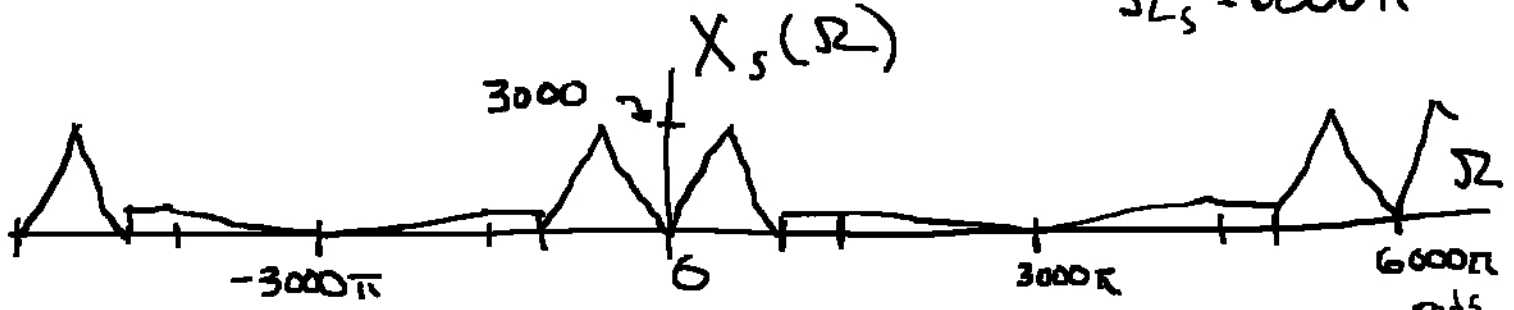
$T = \frac{1}{3000}$ sec

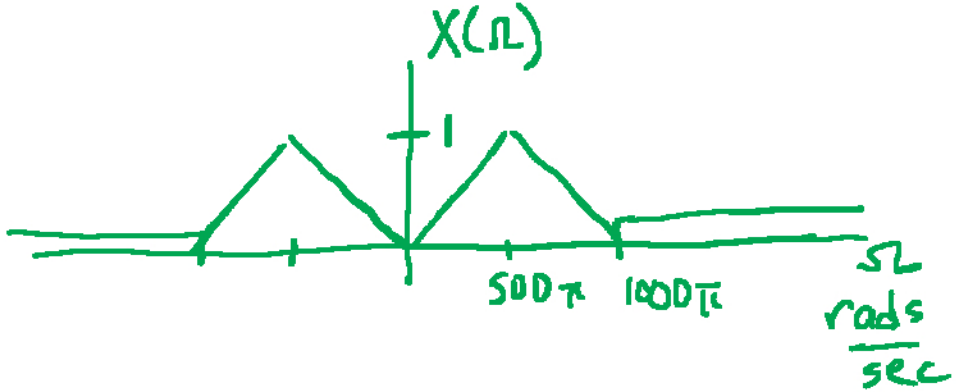
$$X_a(\Omega) = X(\Omega)\, H_a(\Omega)$$



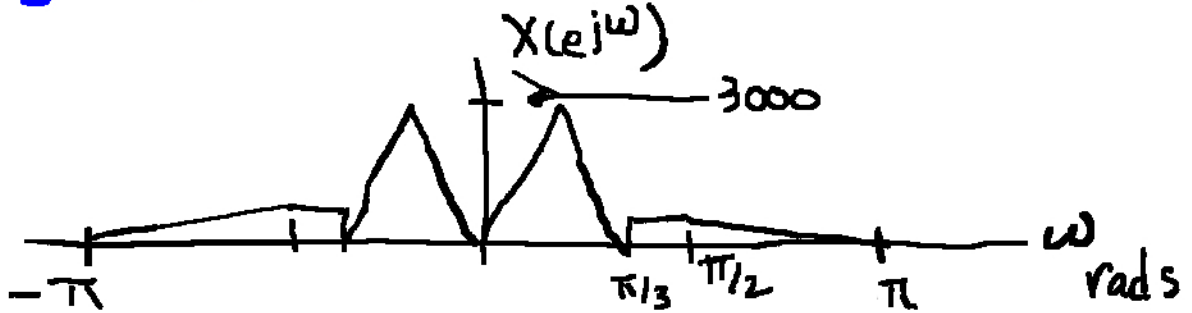
$$x[n] \xrightarrow{FT} X_s(\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_a(\Omega - k\Omega_s)$$

$\Omega_s = 6000\pi$

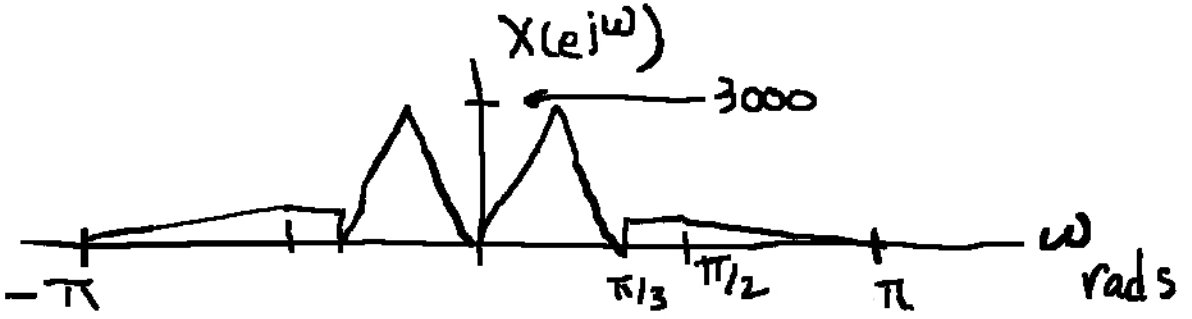




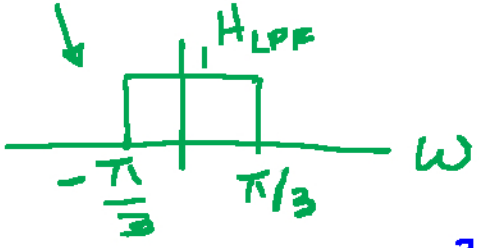
$$x[n] \xrightarrow{DTFT} X(e^{j\omega}) = X_s(\Omega)\big|_{\Omega = \omega/T}$$



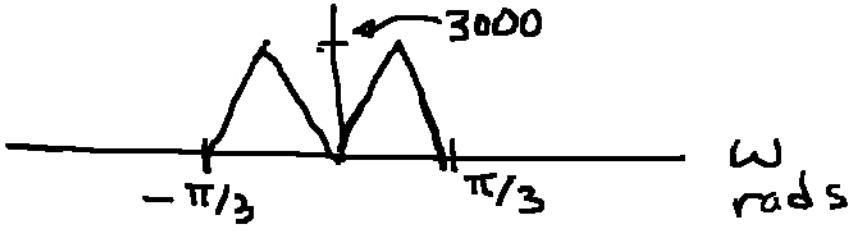






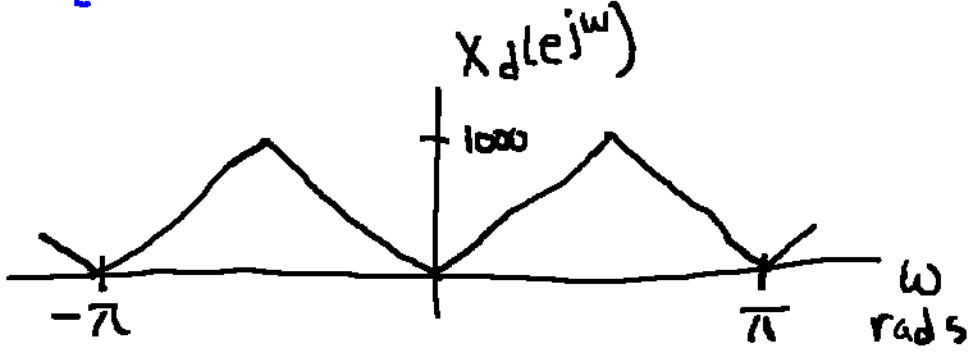


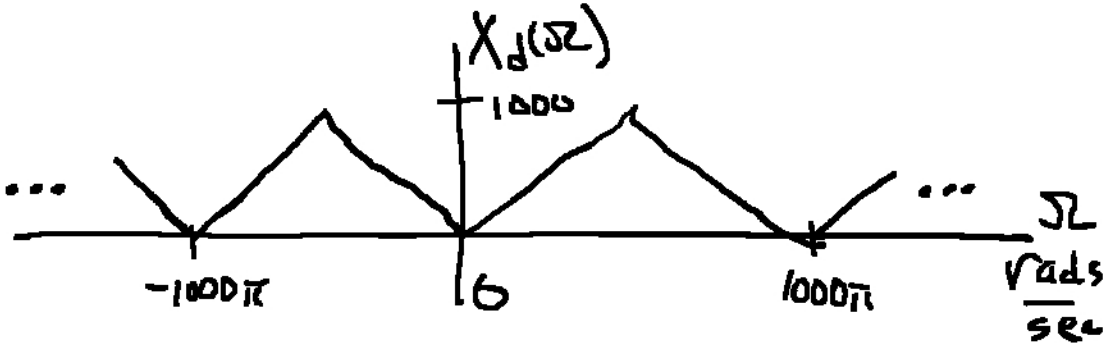
$$X_L(e^{j\omega}) = X(e^{j\omega}) H_{LPF}(e^{j\omega})$$

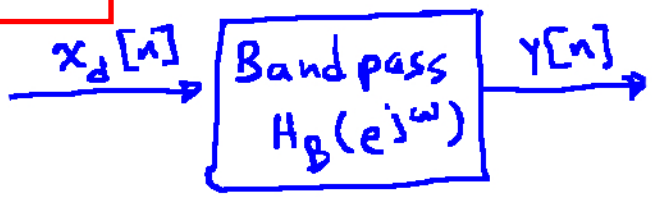


$$X_d(e^{j\omega}) = \frac{1}{3} \sum_{m=0}^{2} X_L(e^{j(\omega - m2\pi)/3})$$

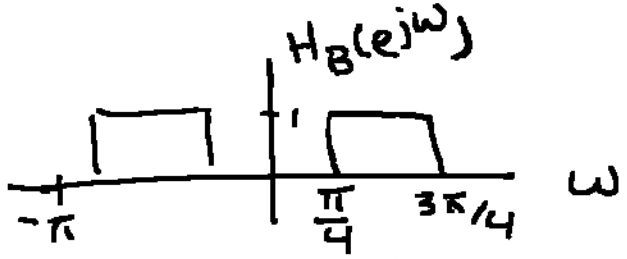
$$X_d(\Omega) = X_d(e^{j\omega}) \Big|_{\omega = \Omega T'} ; T' = \frac{1}{1000}$$

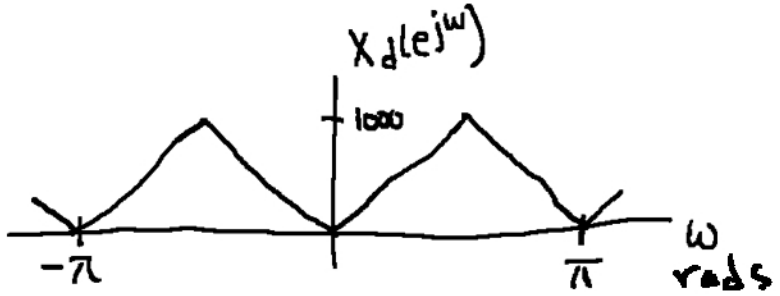


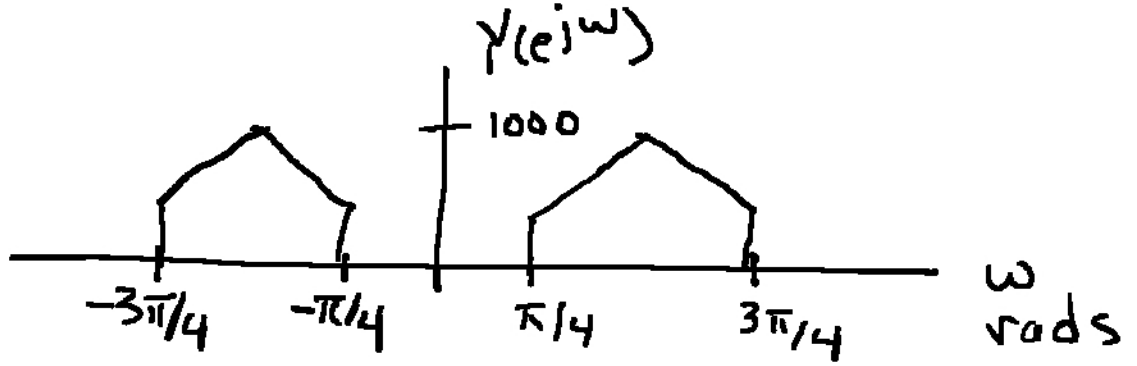




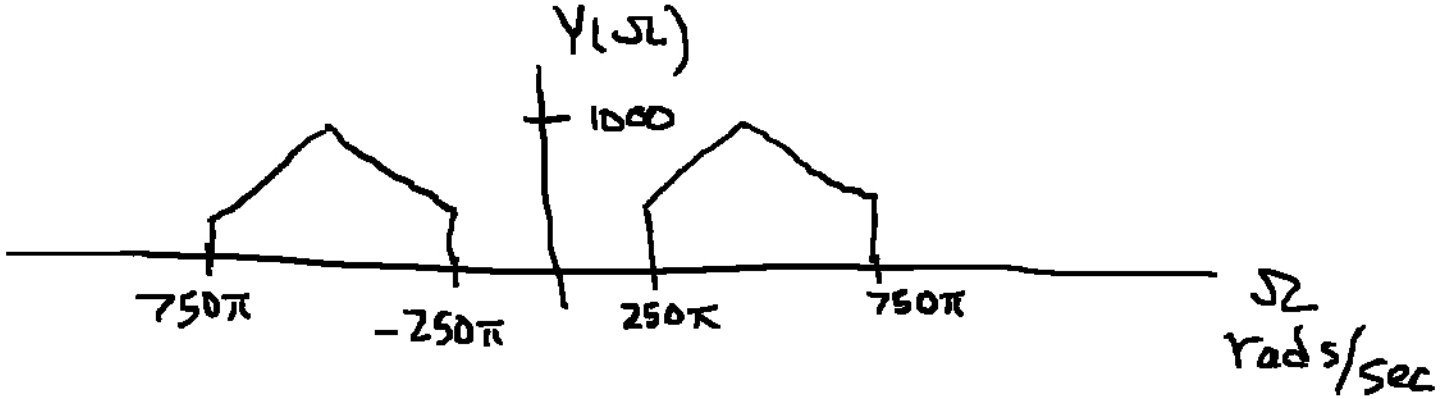
$$Y(e^{j\omega}) = H_B(e^{j\omega})\, X_d(e^{j\omega})$$



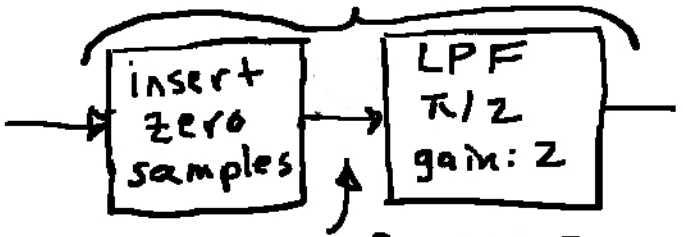




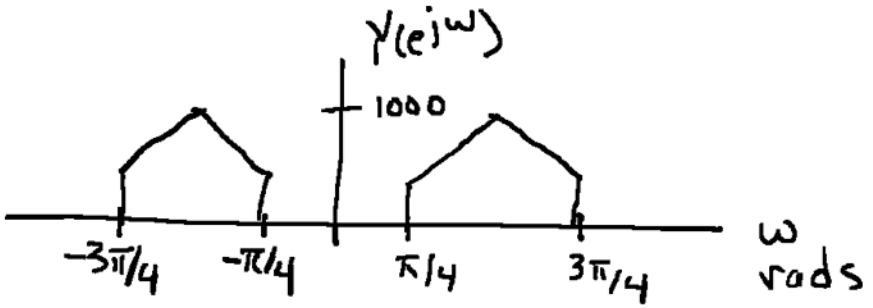
$$Y(\Omega) = Y(e^{j\omega})\Big|_{\omega = \Omega T'} \quad T' = \frac{1}{1000}$$

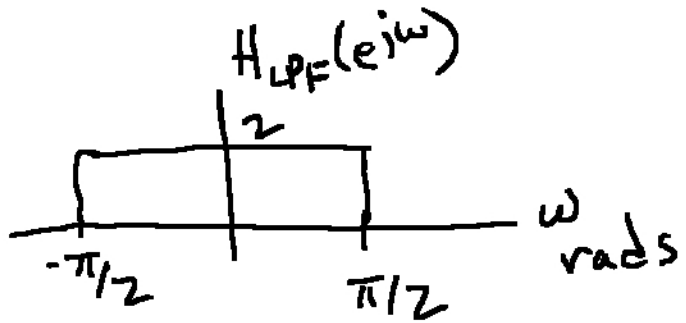




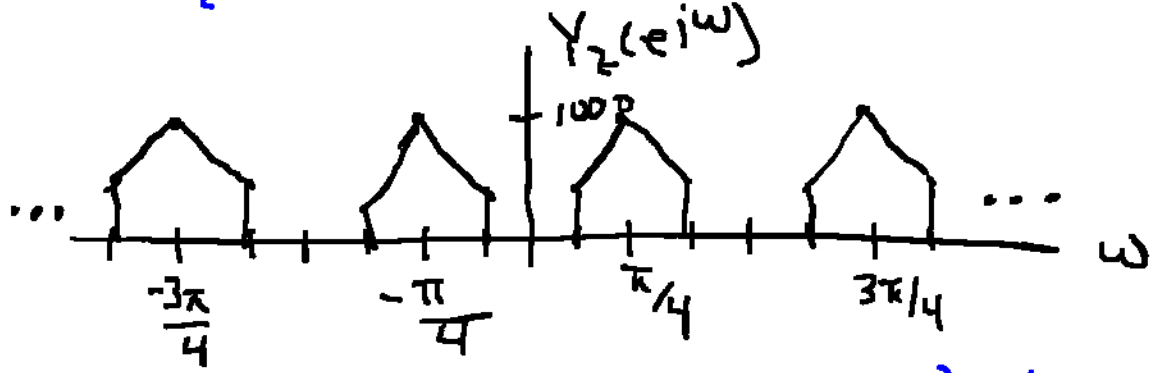


$$y_z[n] = \begin{cases} y[n/2] & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

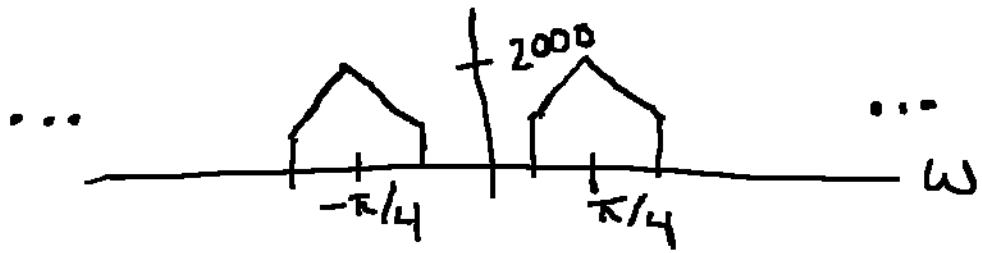




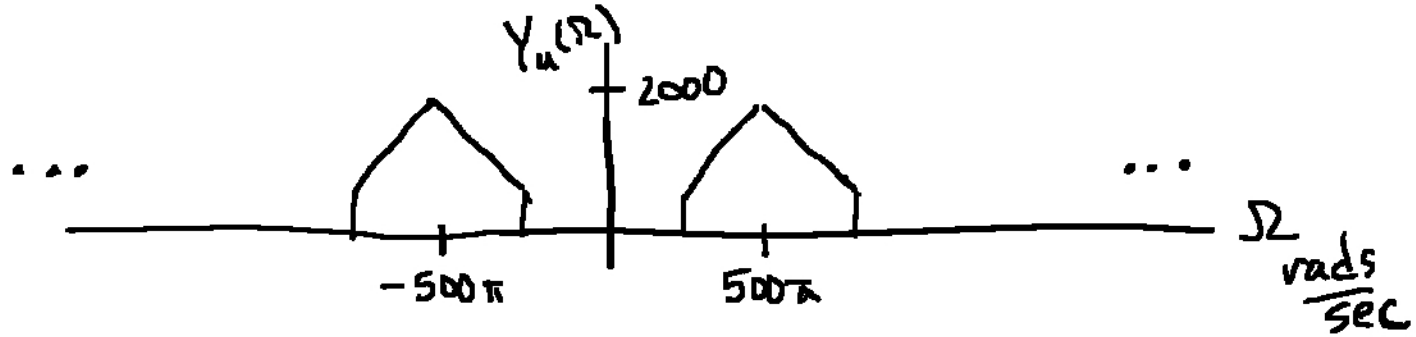
$$Y_z(e^{j\omega}) = Y(e^{j2\omega})$$



$$Y_u(e^{j\omega}) = H_{LPF}(e^{j\omega})\, Y_z(e^{j\omega})$$



$$Y_u(\Omega) = Y_u(e^{j\omega})\big|_{\omega = \Omega T''} \quad ; \quad T'' = \frac{1}{2000}$$



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